MATH406 Notes

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1 Intro

Number theory is mostly about the properties of integers

We will preemptively assume the entirety of set theory and the basics of lin alg.

Textbook: Divisibility

2 Lecture 1

2.1 2.1 - Divisibility

We can add/subtract/multiply integers to get integers (eg prop of field, closed).

Definition: Let a, b be integers, $b \neq 0$. We say that b divides a (denoted $b|a$) if there exists an integer k such that $a = bk$.

Additionally,

- If b does not divide a, we write $b \nmid a$.
- When $d|a$, we say that a is a multiple of d and that d is a divisor (or factor) of a.
- For any nonzero integer d we have $d|0$ since $0 = d \cdot 0$.
- We never consider division by zero, so $0|a$ is not defined.

<u>Convention:</u> When we talk about the set of divisors of a positive integer, we mean the set of positive divisors. For instance, the divisors of 6 are 1,2,3,6 (not -1,-2,-3,-6).

Proposition: Let a, b, c integers. Then $a|b$ and $b|c$ implies $a|c$.

Proof: Since $a|b$, there exists an integer k such that $b = ak$. Since $b|c$, there exists an integer l such that $c = bl$. Thus,

$$\begin{aligned} c &= bl \\ &= (ak)l \\ &= a(kl) \end{aligned}$$

Since kl is an integer under field closure, we have $a|c$. $\square$

Definition: A linear combination of integers is an expression of the form $ax + by$ where x, y are integers.

"A divisor of both a and b is a divisor of any linear combination of a and b."

Proposition: Bilinearity of division; that is, $d|ax + by$ if $d|a$ and $d|b$.

Proof: Since $d|a$, there exists an integer k such that $a = dk$. Since $d|b$, there exists an integer l such that $b = dl$. Thus,

$$\begin{aligned} ax + by &= (dk)x + (dl)y \\ &= d(kx) + d(ly) \\ &= d(kx + ly) \end{aligned}$$

Since $kx + ly$ is an integer under field closure, we have $d|ax + by$. □

Corollary: Suppose a, b, d are integers. Then, if $d|a$ and $d|b$ then $d|(a - b)$ and $d|(a + b)$.

Proof: Let $x = 1$ and $y = -1$ in the previous proposition to get $d|(a - b)$. Let $x = 1$ and $y = 1$ to get $d|(a + b)$. □

2.2 2.2 - Euclid's Theorem

Definition: A <u>prime number</u> is an integer $p \geq 2$ whose only positive divisors are 1 and p itself. An integer $n \geq 2$ that is not prime is called <u>composite</u>.

Theorem (Euclid's Theorem): There are infinitely many prime numbers.

3 Lecture 2

Proof of Euclid's Theorem: Suppose there are finitely many primes $p_1, p_2, \ldots, p_n$. Let

$$Q = p_1 p_2 p_3 \cdots p_n + 1$$

Then Q is either prime or composite. If Q is prime, then Q is a prime not in our list. If Q is composite, then Q has a prime divisor p. Since p divides the product $p_1 p_2 \cdots p_n$, it follows that p does not divide $Q - p_1 p_2 \cdots p_n = 1$. Thus, p is not in our list of all primes. In either case, we have a contradiction. Therefore, there are infinitely many primes. □

3.1 2.4 - The Sieve of Eratosthenes

Theorem (Sieve of Eratosthenes): To find all primes less than or equal to a given integer $n \geq 2$, we can use the following algorithm:

1. Write down all integers from 2 to n.
2. Start with the first number p in the list (which is 2). Circle p and cross out all multiples of p greater than p.
3. Find the next number in the list that is not crossed out. If there is no such number, stop. Otherwise, let p be this new number and repeat step 2.
4. When the algorithm stops, the circled numbers are all the primes less than or equal to n.

Proof: Let p be any prime less than or equal to n. When we reach step 2 with this value of p, we circle it since it has not been crossed out (as it is prime). Thus, all primes less than or equal to n are circled when the algorithm stops. □

4 Lecture 3

4.0.1 2.5 - The Division Algorithm

Consider the grade school long division problem. We want to do, for instance, 95 divided by 7. We can write

$$95 = 7 \cdot 13 + 4$$

Theorem (The Division Algorithm): Let $a, b \in \mathbb{Z}$ and $b > 0$. Then $\exists! q, r \in \mathbb{Z}$ st

$$a = bq + r \quad \text{with} \quad 0 \leq r < b$$

Note that if $b > a$ then $r = a$ and $q = 0$.

Proof: (Existence) Let q be the largest integer such

$$q \leq \frac{a}{b} < q + 1$$

Then $bq \leq a < bq + b$. and so $0 \leq a - bq < b$. Let $r = a - bq$. Then $a = bq + r$ with $0 \leq r < b$.

(Uniqueness) Suppose there exist q', r' such that

$$a = bq' + r' \quad \text{with} \quad 0 \leq r' < b$$

Then

$$\begin{aligned} bq + r &= bq' + r' \\ b(q - q') &= r' - r \end{aligned}$$

If $q \neq q'$, then $|b(q - q')| \geq b$ since $|q - q'| \geq 1$. But $|r' - r| < b$ since both r and r' are between 0 and b. This is a contradiction. Thus, $q = q'$, which implies $r = r'$. $\square$

Example:

1. For $a = 100, b = 9$, we have $100 = 9 \cdot 11 + 1$.
2. For $a = -100, b = 9$, we have $-100 = 9 \cdot (-12) + 8$.

Corollary: $a|b$ iff the remainder when b is divided by a is 0.

Proof: Let $b = aq + r$ by the division algorithm. If $r = 0$, then $b = aq$ and so $a|b$. Conversely, if $a|b$, then $b = ak$ for some integer k. By the uniqueness part of the division algorithm, we must have $q = k$ and $r = 0$. $\square$

4.0.2 2.6 - The GCD

<u>Fact:</u> If you have a nonzero integer a, then it has finite divisors by the well-ordering principle. (eg there are only finitely many positive integers less than or equal to $|a|$, and thus finite candidates for divisors).

Definition (GCD): Let a, b be integers, not both zero. The greatest common divisor of a and b, denoted $\gcd(a, b)$, is the largest positive integer that divides both a and b.

Example:

- $\gcd(12, 15) = 3$

- $\gcd(0, 5) = 5$
- $\gcd(0, 0)$ is not defined.

Definition (Relatively Prime): Two integers are relatively prime if their gcd is 1.

Example:

- $\gcd(8, 15) = 1$ so 8 and 15 are relatively prime.
- $\gcd(9, 28) = 1$ so 9 and 28 are relatively prime.
- $\gcd(0, 1) = 1$ so 0 and 1 are relatively prime.

Corollary: $gcd(p, n) = 1 \iff p \nmid n$.

Proof: If $\gcd(p, n) = 1$, then the only positive divisor of both p and n is 1. Thus, p does not divide n. Conversely, if $p \nmid n$, then the only positive divisor of both p and n is 1, so $\gcd(p, n) = 1$. $\square$

5 Lecture 4

Last time, we defined the GCD.

Proposition (2.10): Let $a, b \in \mathbb{Z}$ and $d = \gcd(a, b)$. Then,

$$\gcd(\frac{a}{d}, \frac{b}{d}) = 1$$

Proof: Recall that the gcd is defined to be at least 1, always. We will prove that

$$\gcd(\frac{a}{d}, \frac{b}{d}) \leq 1$$

To do this, we will show that any common divisor of $\frac{a}{d}$ and $\frac{b}{d}$ is ≤ 1.

Suppose that for some positive integer c, $c|\frac{a}{d}$ and $c|\frac{b}{d}$. By def of divisibility we thus have that $\frac{a}{d} = c \cdot k$, and that $\frac{b}{d} = c \cdot l$ for $k, l \in \mathbb{Z}$.

Then, $a = c \cdot d \cdot k$, and $b = c \cdot d \cdot l$. This shows that cd is a common divisor of a and b. However, d is the gcd of a, b, so it must be that $cd \leq d$. Under $\mathbb{Q}$ field properties there must $\exists!$ inverse to d, $\frac{1}{d}$ and by field closure we can multiply, so $c \leq 1$. $\square$

5.1 2.7 - The Euclidean Algorithm

How can we compute gcds?

One method is to factor each one into primes, take the overlapping subset, and find their product. This is dumb for large integers because factoring large integers is not computationally feasible (for traditional computers)

Lemma: Suppose $a = bq + r$. The set of common divisors of a and b is the same as the set of common divisors of b and r. It thus follows they have the same gcd.

Proof: Suppose $a = bq + r$. Then, a is a linear combination of b, r. Then, any common divisor of b, r must also divide a by a former proposition. Thus, any common divisor of b, r is by definition a divisor of b, but also a divisor of a. Thus, they are also common divisors of a, b.

Similarly, we can rewrite the equation to say that $r = a - bq$, and thus similarly all common divisors of a, b must be common divisors of r, b. So we are done. $\square$

Note: Here we can see the idea for the euclidean algorithm presented. We can reduce the gcd problem from a, b to be r, b, where r is strictly lesser than a. In fact, it's $\leq b$ by definition of division algorithm. Then, we can do this to b, then r again, etc until the numbers are very small.

Example: Let's use this idea to compute $\gcd(315, 220)$. Then,

$$315 = 1 \cdot 220 + 95220 = 2 \cdot 95 + 3095 = 3 \cdot 30 + 530 = 6 \cdot 5 + 0$$

and so applying the lemma, we get that $\gcd(315, 220) = \gcd(220, 95) = \gcd(95, 30) = \gcd(30, 5) = \gcd(5, 0) = 5$.

or the last nonzero remainder, is the gcd.

Theorem (The Euclidean Algorithm): Let $a, b \geq 0 \in \mathbb{Z}, b \neq 0$. Then, $a = bq_1 + r_1$ with $0 \leq r_1 \leq b$. Then, recursively apply the theorem with parameters b, r. The base case applies when the remainder is 0, in which case the divisor is the gcd.

In python,

```
def gcd(a,b):
    if b==0:
        return a
    else:
        return gcd(b, a%b)
```

Proof: This is just repeated application of the lemma. $\square$

5.1.1 2.7.1 - The Extended Euclidean Algorithm

Theorem (2.12): $\exists x, y \in \mathbb{Z}$ st for $a, b \in \mathbb{Z}$ not both zero, $ax + by = \gcd(a, b)$.

Note: They are not unique.

6 Lecture 5

The idea is that we can solve for the remainder. For instance, take the example:

Ex: $\gcd(1239, 735)$ results in

$$\begin{aligned} 1239 &= 1 \cdot 735 + 504 \\ 735 &= 1 \cdot 504 + 231 \\ 504 &= 2 \cdot 231 + 42 \\ 231 &= 5 \cdot 42 + 21 \\ 42 &= 2 \cdot 21 + 0 \end{aligned}$$

But this also gives

$$\begin{aligned} 504 &= 1239 - 1 \cdot 735 \\ 231 &= 735 - 1 \cdot 504 \\ 42 &= 504 - 2 \cdot 231 \\ 21 &= 231 - 5 \cdot 42 \end{aligned}$$

We can substitute these all the way up to get

$$\begin{aligned}
21 &= 231 - 5 \cdot 42 \\
&= 231 - 5 \cdot (504 - 2 \cdot 231) \\
&= 11 \cdot 231 - 5 \cdot 504 \\
&= 11 \cdot (735 - 1 \cdot 504) - 5 \cdot 504 \\
&= 11 \cdot 735 - 16 \cdot 504 \\
&= 11 \cdot 735 - 16 \cdot (1239 - 1 \cdot 735) \\
&= 27 \cdot 735 - 16 \cdot 1239
\end{aligned}$$

This gives us $x = -16$ and $y = 27$ such that $1239 \cdot (-16) + 735 \cdot 27 = 21$.

Theorem (The Extended Euclidean Algorithm): Let $a, b \in \mathbb{Z}$ not both zero. Then, we can find $x, y \in \mathbb{Z}$ such that $ax + by = \gcd(a, b)$. Do so by:

- Apply the Euclidean algorithm to find the gcd and all the remainders.
- Rewrite each remainder as a linear combination of a and b by substituting the previous remainders.
- Continue substituting until you express the gcd as a linear combination of a and b.

In python,

```
def extended_gcd(a,b):
    if b==0:
        return (a,1,0)
    else:
        d,x1,y1 = extended_gcd(b, a%b)
        x = y1
        y = x1 - (a//b)*y1
        return (d,x,y)
```

Proof: This is just the algorithm described above. The correctness follows from the Euclidean algorithm and properties of linear combinations. $\square$

<u>Example:</u> Find x, y such that $19x + 7y = 1$.

We apply the extended euclidean algorithm:

$$\begin{aligned}
19 &= 2 \cdot 7 + 5 \\
7 &= 1 \cdot 5 + 2 \\
5 &= 2 \cdot 2 + 1 \\
2 &= 2 \cdot 1 + 0
\end{aligned}$$

Then, in reverse,

$$\begin{aligned}
1 &= 5 - 2 \cdot 2 \\
&= 5 - 2 \cdot (7 - 1 \cdot 5) \\
&= 3 \cdot 5 - 2 \cdot 7 \\
&= 3 \cdot (19 - 2 \cdot 7) - 2 \cdot 7 \\
&= 3 \cdot 19 - 8 \cdot 7
\end{aligned}$$

Some applications:

Proposition (2.16): Let $a, b, c \in \mathbb{Z}$ st $a \neq 0, \gcd(a, b) = 1$. Then if $a|bc$, then $a|c$.

The idea is that if a, b are coprime, then a has no common factors with b. Thus, if a divides bc, it must be that a divides c.

Proof: Since $\gcd(a, b) = 1$, there exist $x, y \in \mathbb{Z}$ such that $ax + by = 1$. Multiplying both sides by c gives $acx + bcy = c$. Since $a|acx$ and $a|bcy$ (since $a|bc$), it follows that $a|c$. $\square$

Corollary: Let $p \in \mathbb{Z}$ prime, and $a, b \in \mathbb{Z}$. Then if $p|ab$, then $p|a$ or $p|b$.

Proof: If $p|a$, we are done. Otherwise, $\gcd(p, a) = 1$ since p is prime and does not divide a. Thus, by the previous proposition, $p|b$. $\square$

Corollary (2.14): Let $a, b, d \in \mathbb{Z}$ st $d = \gcd(a, b)$. For $e \in \mathbb{Z}$ if $e|a, e|b$, then $e|d$.

Proof: Since d is a linear combination of a and b, we have that $d = ax + by$ for some integers x, y. Since $e|a$ and $e|b$, it follows that $e|ax$ and $e|by$. Thus, $e|d$. $\square$

6.1 2.9 - Fermat and Mersenne Primes

We know that there are infinitely many primes by Euclid's theorem. However, are there infinitely many primes of certain forms? And can we use this to generate primes? Useful for cryptography, etc.

Definition (Mersenne Number): The nth Mersenne number is defined as $M_n = 2^n - 1$ for $n \geq 1$.

Proposition: If $a|n$, then $M_a|M_n$. If n is composite, then M_n is composite.

Proof: Suppose $n = ab$ for integers $a, b > 1$. Then,

$$\begin{aligned} M_n &= 2^{ab} - 1 \\ &= (2^a)^b - 1^b \\ &= (2^a - 1)\left((2^a)^{b-1} + (2^a)^{b-2} + \cdots + 1\right) \end{aligned}$$

Since $a, b > 1$, it follows that $2^a - 1 > 1$ and the second factor is also greater than 1. Thus, M_n is composite. $\square$

<u>Example:</u> $M_10 = 2^{10} - 1 = 1023$ is composite. We know that $2|10$ so $M_2|M_{10}$. In fact, $M_2 = 3$ and $1023 = 3 \cdot 341$. Also, $5|10$ so $M_5|M_{10}$. In fact, $M_5 = 31$ and $1023 = 31 \cdot 33$.

<u>Note:</u> If p is prime, is M_p prime? For some, yes, i.e. 13, 17, 19. However M_{11} isn't prime for instance.

Definition (Fermat Number): The nth Fermat number is defined as $F_n = 2^{2^n} + 1$ for $n \geq 0$.

<u>Note:</u> He initially looked at $2^m + 1$ but found that unless m is a power of 2, $2^m + 1$ is composite. For instance, $2^6 + 1 = 65$ is composite.

Proposition: For $2^m + 1$, if m is not a power of 2, then $2^m + 1$ is composite. If $m = 2^n$, then $2^m + 1$ is prime for $n = 0, 1, 2, 3$ but composite for $n = 4$.

Proof: Suppose m is not a power of 2. Recall that if k is odd, then $x^k + 1$ is divisible by $x + 1$. Since m is

not a power of 2, then we can write $m = 2^n \cdot k$ for some odd integer $k > 1$. Then,

$$\begin{aligned} 2^m + 1 &= 2^{2^n \cdot k} + 1 \\ &= (2^{2^n})^k + 1^k \\ &= (2^{2^n} + 1)\left((2^{2^n})^{k-1} - (2^{2^n})^{k-2} + \cdots - 1\right) \end{aligned}$$

Since $k > 1$, it follows that $2^{2^n} + 1 > 1$ and the second factor is also greater than 1. Thus, $2^m + 1$ is composite. $\square$

7 Lecture 6

Proposition: If $M_p = 2^p - 1$ is prime, then p is prime. Such an M_p is called a Mersenne prime.

<u>Note:</u> There are only 52 known Mersenne primes! And it's not known if there are infinitely many.

It turns out that F_5, F_6, F_7 are all composite. In fact, only F_0 to F_4 are known to be prime. It is not known if there are infinitely many Fermat primes.

8 Lecture 7

Chapter 3 now, Linear Diophantine Equations.

8.1 3.1 - Linear Diophantine Equations

Take $ax + by = c$ for $a, b, c \in \mathbb{Z}$. We are interested in integer pairs (x, y) that satisfy this linear diophantine equation.

The goal will be to describe <u>all</u> such solutions (x, y)

<u>Example:</u> Find integers x, y such that $19x + 7y = 1$.

By the extended euclidean algorithm,

$$\begin{aligned} 19 &= 2 \cdot 7 + 5 \\ 7 &= 1 \cdot 5 + 2 \\ 5 &= 2 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 + 0 \end{aligned}$$

Then, in reverse,

$$\begin{aligned} 1 &= 5 - 2 \cdot 2 \\ &= 5 - 2 \cdot (7 - 1 \cdot 5) \\ &= 3 \cdot 5 - 2 \cdot 7 \\ &= 3 \cdot (19 - 2 \cdot 7) - 2 \cdot 7 \\ &= 3 \cdot 19 - 8 \cdot 7 \end{aligned}$$

Then, one solution is $3, -8$. There are infinitely many solutions, since we can add $7k$ to x and subtract $19k$ from y for any integer k to get another solution. So, the set of all solutions is $\{(3 + 7k, -8 - 19k) : k \in \mathbb{Z}\}$.

<u>Example:</u> Find integers x, y such that $19x + 7y = 6$.

We do know that

$$\begin{aligned} 1 &= 3 \cdot 19 - 8 \cdot 7 \\ 6 &= 18 \cdot 19 - 48 \cdot 7 \end{aligned}$$

Then, we can similarly add $7k$ to x and subtract $19k$ from y for any integer k to get another solution. So, the set of all solutions is $\{(18 + 7k, -48 - 19k) : k \in \mathbb{Z}\}$.

<u>Example:</u> Find integers x, y such that $6x + 15y = 10$.

There are no integer solutions since $\gcd(6, 15) = 3$ does not divide 10. Generally if $\gcd(a, b)$ does not divide c, then there are no integer solutions to $ax + by = c$.

Proof: Let $d = \gcd(a, b)$. Then $d|a, d|b$ and $d|ax + by$ by bilinearity of divisibility. However, $d \nmid c$ by assumption, so there are no integer solutions to $ax + by = c$. $\square$

Theorem (3.1): Assume $a, b, c \in \mathbb{Z}$, and at least one of $a, b \neq 0$, and let $d = \gcd(a, b)$. Then,

(1) The equation $ax + by = c$ has integer solutions iff $d|c$.

(2) Additionally, if it has an integer solution, it has infinitely many integer solutions. In particular, if (x_0, y_0) is one integer solution, then the set of all integer solutions is given by

$$\{(x_0 + \frac{b}{d}k, y_0 - \frac{a}{d}k) : k \in \mathbb{Z}\}$$

Proof:

(1) $\Rightarrow$: Let $ax + by = c$ have integer solutions. Then, since $d|a, d|b$, we have that $d|ax + by$ by bilinearity of divisibility. Thus, $d|c$.

$\Leftarrow$ Suppose $d|c$. THen, $c = d \cdot e$ for some integer e. Since d is a linear combination of a and b, we have that $d = ax_0 + by_0$ for some integers x_0, y_0. Thus,

$$\begin{aligned} c &= d \cdot e \\ &= (ax_0 + by_0)e \\ &= a(ex_0) + b(ey_0) \end{aligned}$$

So, (ex_0, ey_0) is an integer solution to $ax + by = c$.

(2) (This is my own proof) First, $\frac{b}{d}, \frac{a}{d} \in \mathbb{Z}$ by proposition 2.10. Now, let (x_0, y_0) be one integer solution to $ax + by = c$. Then, for any integer k, we have

$$\begin{aligned} a(x_0 + \frac{b}{d}k) + b(y_0 - \frac{a}{d}k) &= ax_0 + by_0 + ab\frac{k}{d} - ab\frac{k}{d} \\ &= c \end{aligned}$$

Thus, $(x_0 + \frac{b}{d}k, y_0 - \frac{a}{d}k)$ is an integer solution for any integer k.

$\square$

Definition (Homogeneous LDE): A linear diophantine equation of the form $ax + by = 0$ is called a homogeneous linear diophantine equation.

Lemma: Suppose a, b are not both 0. Then

(1) If $\gcd(a,b)=1$, every solution to $ax+by=0$ is of the form $(bk,-ak)$ for some integer k.

(2) In general, if $d=\gcd(a,b)$, every solution to $ax+by=0$ is of the form $(\frac{b}{d}k,-\frac{a}{d}k)$ for some integer k.

Proof:

(1) Let a,b not both 0. Suppose $\gcd(a,b)=1$. Let (x,y) be a solution to $ax+by=0$. Then, $ax=-by$ and so $a|by$. Since $\gcd(a,b)=1$, it follows that $a|y$. Thus, $y=ak$ for some integer k. Substituting back gives $ax=-bak$, so $x=-bk$. Thus, every solution is of the form $(bk,-ak)$ for some integer k.

(2) Let a,b not both 0. Let $d=\gcd(a,b)$. Let (x,y) be a solution to $ax+by=0$. Then, $ax=-by$ and so $a|by$. Since $\gcd(a,b)=d$, it follows that $\frac{a}{d}|\frac{b}{d}y$. Since $\gcd(\frac{a}{d},\frac{b}{d})=1$ by proposition 2.10, it follows that $\frac{a}{d}|y$. Thus, $y=\frac{a}{d}k$ for some integer k. Substituting back gives $ax=-b\frac{a}{d}k$, so $x=-\frac{b}{d}k$. Thus, every solution is of the form $(\frac{b}{d}k,-\frac{a}{d}k)$ for some integer k.

9 Lecture 8

Theorem (3.1): Let $d=\gcd(a,b)$. Then

1. $ax+by=c$ has solutions iff $d|c$.

2. If (x_0,y_0) is one solution to $ax+by=c$, then the solution set is given by

$$x=x_0+\frac{b}{d}t, y=y_0-\frac{a}{d}t$$

where t is an integer

We proved this last time, along with a related lemma. I will now write Prof's proof of part 2.

Proof: (of part 2) Suppose (x_0,y_0) is some solution for $ax+by=c$. Then, we should show that the above format gives solutions, and additionally that they are all of the solutions. So

- Verify it is a solution:

$$\begin{aligned} a(x_0+\frac{b}{d}t)+b(y_0+\frac{a}{d}t) &= ax_0+\frac{ab}{d}t+by_0-\frac{ab}{d}t \\ &= ax_0+by_0 \end{aligned}$$

which is a solution by assumption.

- Verify it is all the solutions: Let (x,y) be any solution to $ax+by=c$. Then,

$$\begin{aligned} a(x-x_0)+b(y-y_0) &= ax+by-(ax_0+by_0) \\ &= c-c \\ &= 0 \end{aligned}$$

Thus, $(x-x_0,y-y_0)$ is a solution to the homogeneous linear diophantine equation $ax+by=0$. By the previous lemma, there exists an integer t such that

$$(x-x_0,y-y_0)=(\frac{b}{d}t,-\frac{a}{d}t)$$

Thus,

$$(x,y)=(x_0+\frac{b}{d}t,y_0-\frac{a}{d}t)$$

which is of the desired form.

$\square$

Note: Thus, to solve a linear diophantine equation

$$ax + by = c$$

, first ensure that $d = \gcd(a, b)$ divides c. If not, there are no solutions. If it does, then find one solution (x_0, y_0) using the extended euclidean algorithm. Then, the set of all solutions is given by

$$\{(x_0 + \frac{b}{d}t, y_0 - \frac{a}{d}t) : t \in \mathbb{Z}\}$$

Example: Solve $12x + 45y = 30$. Then, $\gcd(12, 45) = 3, 3|30$ so there are solutions

$$\begin{aligned} 45 &= 12(3) + 9 \\ 12 &= 9(1) + 3 \\ 9 &= 3(3) + 0 \end{aligned}$$

going back up we have

$$\begin{aligned} 3 &= 12 - 9(1) \\ &= 12 - (45 - 12(3)) \\ &= 4 \cdot 12 - 45 \end{aligned}$$

we can multiply by 10 to get $30 = 40 \cdot 12 - 10 \cdot 45$. Thus, one solution is $(40, -10)$. The set of all solutions is given by

$$\{(40 + 15t, -10 - 4t) : t \in \mathbb{Z}\}$$

9.1 3.2 - Postage Stamp Problem

Problem: You have an unlimited supply of 3 cent and 5 cent stamps. What amounts of postage can you make? Generalize this to a cent and b cent stamps, frequently where a, b are coprime.

The answer is obviously intuitively

$$ax + by = c$$

for some nonnegative integers x, y. We want to find the largest c such that there are no nonnegative integer solutions to this equation. Also, what are all of the values of c such that there are no nonnegative integer solutions to this equation?

> Professor gives an example for postage stamps with values of 3 and 5 here, which is a trivially solved problem. The idea is that if you can find 3 consecutive values of c that can be achieved, then all values of c greater than the largest of those 3 must also be achievable. This is because you can add 3 to any of those 3 values to get the next value, and so on. Thus, you just need to find the largest c such that there are no 3 consecutive values of c that can be achieved.
>
> In general, we can intuit from this that for the smallest of a, b, if we can find a consecutive values of c that can be achieved, then all values of c greater than the largest of those a must also be achievable. Thus, you just need to find the largest c such that there are no a consecutive values of c that can be achieved.

Example: Suppose in some fucked up version of football the only ways to score are with 3 or 7 points. What scores can be achieved?

Remember you just need 3 in a row that can be achieved and everything after is doable.

Score	Achievable?
0	Yes
1	No
2	No
3	Yes
4	No
5	No
6	Yes
7	Yes
8	No
9	Yes
10	Yes
11	No
12	Yes
13	Yes
14	Yes

and so on. We can see that the largest score that cannot be achieved is 11, and all scores greater than 11 can be achieved.

Proposition (3.3): Suppose $a > 1, b > 1$ are coprime. Then there are no nonnegative integers x, y such that $ax + by = ab - a - b$.

<u>Note:</u> The general idea behind this proof is that when you plug in any arbitrary integer solution to $ax+by = ab - a - b$, you get that $ax_0 + by_0 = 1$. Since $a, b > 1$, this forces x_0, y_0 to be nonnegative and sum to at most 1, none of which can satisfy the equation.

Proof: Since a, b are coprime, there exist integers by a previous theorem that $ax_0 + by_0 = 1$.

We know that $(ab-a-b)x_0, (ab-a-b)y_0$ is a solution to $ax+by = ab-a-b$. Since $a, b > 1, ab-a-b > 0$, and so for both to be positive we must have $x_0, y_0 \geq 0$.

Thus, $ax_0 + by_0 \geq 0$. However, since x_0, y_0 are integers, they must sum to at most 1, otherwise $ax_0 + by_0 \geq 2a + 2b > 1$. Thus, $0 \leq x_0 + y_0 \leq 1$. By integer properties, this means that either $x_0 = 0, y_0 = 0$ or one of them is 1 and the other is 0.

Then, if $x_0 = 0, y_0 = 0$, then $ax_0 + by_0 = 0 \neq 1$. If $x_0 = 1, y_0 = 0$, then $ax_0 + by_0 = a > 1$. If $x_0 = 0, y_0 = 1$, then $ax_0 + by_0 = b > 1$. But we need it to be identically 1 by assumption, a contradiction. $\square$

The professor gives a proof that requires memorizing some answer for x_0, y_0, which is why I derived my own.

Proof: (by the professor) Consider the Diophantine Equation

$$ax + by = ab - a - b$$

Notice that $x_0 = -1, y_0 = a - 1$ is a solution, as

$$\begin{aligned} a(-1) + b(a-1) &= -a + ab - b \\ &= ab - a - b \end{aligned}$$

Thus, the general solution of the Diophantine Equation is $x = -1 + bt, y = a - 1 - at$ for some integer t.

WLOG, suppose (x, y) is a solution such that $x \geq 0$. Since $x \geq 0$, we have that $-1 + bt \geq 0$. This

implies that $t > 0$, an integer, so $t \geq 1$. Then, since $y = a-1-at$, we have that $y \leq a-1-a = -1 < 0$. Thus, if $x \geq 0$, then $y < 0$. Similarly, if $y \geq 0$, then $x < 0$. Thus, there are no nonnegative integer solutions to the equation. $\square$

10 Lecture 9

Recall: The postage stamp problem asks, given a, b coprime positive integers, what is the largest integer c such that there are no nonnegative integer solutions to $ax + by = c$?

Proposition (3.4): Let a, b be coprime positive integers. If $n \geq ab - a - b + 1$ then there are nonnegative integers x, y, such that $ax + by = n$.

Proof: Let (x_0, y_0) be an integer solution to $ax + by = n$, which exists because $\gcd(a, b) = 1$ divides n. Then, the set of all integer solutions is given by

$$\{(x_0 + bk, y_0 - ak) : k \in \mathbb{Z}\}$$

Divide y_0 by a to get $y_0 = qa + r$ for some integer q and $0 \leq r < a$ by the division algorithm. Consider the solution (x_1, y_1) corresponding to $k = q$, which is given by

$$\begin{aligned}(x_1, y_1) &= (x_0 + bq, y_0 - ak) \\ &= (x_0 + bq, r)\end{aligned}$$

Thus, $0 \leq y_1 < a$. We need to show that $x_1 \geq 0$. Note that we have that $n \geq ab - a - b - 1$, and also that $y_1 \leq a - 1$ because $y_1 = r < a$. This implies $-y_1 \geq -(a - 1)$. Since $ax_1 + by_1 = n$, we have

$$\begin{aligned}x_1 &= \frac{n - by_1}{a} \\ &\geq \frac{ab - a - b + 1 - b(a - 1)}{a} \\ &= \frac{ab - a - b + 1 - ab + b}{a} \\ &= \frac{1 - a}{a} \\ &= -1 + \frac{1}{a}\end{aligned}$$

But because x_1 is an integer, it must be that $x_1 \geq 0$. Thus, there are nonnegative integer solutions to $ax + by = n$. $\square$

Corollary: From the above two propositions, the largest integer c such that there are no nonnegative integer solutions to $ax + by = c$ is $ab - a - b$. That is, for all $n \geq ab - a - b + 1$, there are nonnegative integer solutions to $ax + by = n$, and for $n = ab - a - b$, there are no nonnegative integer solutions to $ax + by = n$.

Proof: Compose the previous 2 proofs.

Example: The 2026 Postage Stamp Problem uses values of:

Stamp	Stamp Value
Regular	78
Postcard	61

First, we must have that they are coprime. Since $78 = 2 \cdot 3 \cdot 13$ and 61 is prime, they are coprime.

Then, under this problem, the largest amount of postage that cannot be made is $78 \cdot 61 - 78 - 61 = 4619$. For all amounts of postage greater than 4619, there are nonnegative integer solutions to $78x + 61y = n$.

Example: Find the smallest for stamp values of $6, 8$. These two are not coprime, so we will divide the equation by their gcd (2). Then, we find the largest amount of postage that cannot be made with stamps of values $3, 4$, which is $3 \cdot 4 - 3 - 4 = 5$. Then, we multiply this by the gcd (2) to get 10 as the largest amount of postage that cannot be made with stamps of values $6, 8$. For all amounts of postage greater than 10, there are nonnegative integer solutions to $6x + 8y = n$.

10.1 QUIZ 1 CUTOFF HERE

10.2 Appendix A.3 - Well-Ordering Principle and Mathematical Induction

Theorem (Well-Ordering Principle): Every nonempty subset of $\mathbb{N}$ (the positive integers) has a least element.

Note: This only applies to the naturals and not the integers, since the integers can be negative and thus have no least element. Most notably the proof of the well-ordering principle uses the principle of succession, which is a property of the naturals specifically due to Peano succession axioms. You can also prove this utilizing the completeness of the reals, but once again this requires the existence of integers specifically in a half open interval.

Specifically it requires the completeness property, showing that all bottom-bounded subsets of the reals have a least upper bound, and because of the transitivity of $\subseteq$, for $A \subseteq \mathbb{N}$, A has an infinum called a, and then $(n-1, n]$ must contain an element of A for all $n \geq a$, so there must be some least element of A.

Note: We will use this to give a non-constructive proof of the following theorem:

Theorem (2.12): Let a, b be integers, not both 0 and let $d = \gcd(a, b)$. Then, there exist integers x, y such that $ax + by = d$.

Note: Last time, we proved this using the Extended Euclidean Algorithm, which is a constructive proof. This time, we will give a non-constructive proof using the Well-Ordering Principle.

Proof: Let $S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\}$ be the set of all positive linear combinations of a and b. Since a, b are not both 0, there exists some positive linear combination of a and b, so S is nonempty. By the Well-Ordering Principle, S has a least element, say d. We will show that $d = \gcd(a, b)$.

First, we show that $d|a$ and $d|b$. By the division algorithm, we can write $a = dq + r$ for some integers q, r with $0 \leq r < d$. Then,

$$\begin{aligned} r &= a - dq \\ &= a - q(ax_0 + by_0) \\ &= a(1 - qx_0) + b(-qy_0) \end{aligned}$$

where x_0, y_0 are integers such that $d = ax_0 + by_0$. Thus, r is a linear combination of a and b. If $r > 0$, then $r \in S$ and $r < d$, contradicting the minimality of d. Thus, $r = 0$ and so $d|a$. Similarly, we can show that $d|b$.

> Professor starts a contradiction by stating suppose $r \neq 0$ and then doing the above, instead of taking cases.

Next, we show that if e is any common divisor of a and b, then $e|d$. Since $e|a$ and $e|b$, it follows that $e|ax+by$ for any integers x, y. In particular, since $d = ax_0 + by_0$, it follows that $e|d$. Since $d > 0$, $e \leq d$. Thus, d is the greatest common divisor of a and b. $\square$

Note: This argument shows that $\gcd(a, b)$ is the least positive linear combination of a and b.

Remark: Let $d = \gcd(a, b)$.

11 Lecture 10

<u>Induction:</u> Let $P(n)$ be a statement about a positive integer n. For instance, n is a product of primes. Then, if $P(1)$ is true, and for all $n \geq 1$, if $P(n)$ is true, then $P(n+1)$ is true, then $P(n)$ is true for all positive integers n.

The induction scheme expressed in the above is called weak induction. There is also strong induction, which states that if $P(1)$ is true, and for all $n \geq 1$, if $P(1), P(2), \ldots, P(n)$ are all true, then $P(n+1)$ is true, then $P(n)$ is true for all positive integers n.

Note that this means that for strong induction, we only have to prove $n = 1$, but we can *assume* that $P(1), P(2), \ldots, P(n)$ are all true to prove $P(n+1)$. For weak induction, we have to prove $n = 1$ and then for each $n \geq 1$, we have to prove that if $P(n)$ is true, then $P(n+1)$ is true. Thus, strong induction is often easier to use than weak induction.

> I think we are allowed to do so because of the well-ordering principle, which states that every nonempty subset of the positive integers has a least element. If there is some n such that $P(n)$ is false, then there is a least such n. Since $P(1)$ is true, $n > 1$. Since n is the least such integer, $P(1), P(2), \ldots, P(n-1)$ are all true. Thus, by the strong induction hypothesis, $P(n)$ is true, a contradiction. Thus, there are no such integers n such that $P(n)$ is false, and so $P(n)$ is true for all positive integers n.

Additionally, we can express them from a logic perspective:

- $P(1) \wedge \forall n \geq 1, P(n) \to P(n+1) \to \forall n \geq 1, P(n)$
- $P(1) \wedge \forall n \geq 1, (P(1) \wedge P(2) \wedge \cdots \wedge P(n)) \to P(n+1) \to \forall n \geq 1, P(n)$

<u>Example:</u> Let's prove that $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for all $n \geq 1$. That is,

$$\sum_{k=1}^{n} (2k-1) = n^2$$

Proof: We will use weak induction. For the base case, $n = 1$, we have $1 = 1^2$, so the base case holds. Now, suppose that for some $n \geq 1$, we have $\sum_{k=1}^{n}(2k-1) = n^2$. Then,

$$\begin{aligned}\sum_{k=1}^{n+1}(2k-1) &= \sum_{k=1}^{n}(2k-1) + (2(n+1)-1) \\ &= n^2 + 2n + 1 \\ &= (n+1)^2\end{aligned}$$

Thus, by the principle of weak induction, we have that $\sum_{k=1}^{n}(2k-1) = n^2$ for all $n \geq 1$.

<u>Example:</u> Prove that $n! \geq n \cdot 2^n$ for all $n \geq 6$.

Proof: By induction,

- <u>Base:</u> $n = 6$, $6! = 720 \geq 6 \cdot 2^6 = 384$, so the base case holds.
- <u>Induction Step:</u> Suppose that for some $n \geq 6$, we have $n! \geq n \cdot 2^n$. Then,

$$\begin{aligned}(n+1)! &= (n+1)n! \\ &\geq (n+1)n \cdot 2^n \\ &\geq (n+1) \cdot 2^{n+1}\end{aligned}$$

Thus, by the principle of weak induction, we have that $n! \geq n \cdot 2^n$ for all $n \geq 6$. $\square$

Example: Prove that $3|(n^3 - n)$ for all $n \geq 0$

Proof: By induction,

- Base: $n = 0$, $3|0^3 - 0 = 0$, so the base case holds.
- Induction Step: Suppose that for some $n \geq 0$, we have $3|(n^3 - n)$. Then,

$$\begin{aligned}(n+1)^3 - (n+1) &= n^3 + 3n^2 + 3n + 1 - n - 1 \\ &= n^3 - n + 3n^2 + 3n \\ &= (n^3 - n) + 3(n^2 + n)\end{aligned}$$

  Since $3|(n^3 - n)$ by the induction hypothesis, and $3|(3(n^2 + n))$ by bilinearity of divisibility, it follows that $3|((n^3 - n) + 3(n^2 + n))$. Thus, by the principle of weak induction, we have that $3|(n^3 - n)$ for all $n \geq 0$. □

Note: Replace 3 with any prime p to get that $p|(n^p - n)$ for all $n \geq 0$. This is a special case of Fermat's Little Theorem, which states that if p is prime and a is an integer not divisible by p, then $a^{p-1} \equiv 1 \mod p$.

Lemma (4.3): Every integer $n \geq 2$ can be written as a product of primes

$$n = p_1 p_2 \cdots p_k$$

Proof: We will use strong induction on n.

- Base: For $n = 2$, $2 = 2$ is a product of primes, so the base case holds.
- Inductive Step: Now assume every integer k with $2 \leq k \leq n$ can be written as a product of primes. We want to show that $n + 1$ can be written as a product of primes. If $n + 1$ is prime, then we are done. If $n+1$ is composite, then there exist integers a, b such that $2 \leq a, b \leq n$ and $n+1 = ab$. By the induction hypothesis, we can write $a = p_1 p_2 \cdots p_k$ and $b = q_1 q_2 \cdots q_m$ for some primes p_i, q_j. Thus,

$$\begin{aligned}n + 1 &= ab \\ &= (p_1 p_2 \cdots p_k)(q_1 q_2 \cdots q_m) \\ &= p_1 p_2 \cdots p_k q_1 q_2 \cdots q_m\end{aligned}$$

  which is a product of primes. Thus, by the principle of strong induction, every integer $n \geq 2$ can be written as a product of primes. □

12 Lecture 11

Recall last time we did 3 main things; the well ordering principle, induction, and strong induction.

All 3 of the above are equivalent to each other. This potentially means that you ould rewrite one proof using the others!

12.1 4.1 0 The Starting Point (of Unique Factorizations)

Theorem (4.1): Let p be a prime and a, b be integers such that $p|ab$. Then, $p|a$ or $p|b$.

Proof: We proved this above in the euclidean algorithm process

Corollary (4.2): Let p be a prime and $a_1, a_2, \ldots, a_k$ be integers such that $p|a_1 a_2 \cdots a_k$. Then, there exists some i such that $p|a_i$.

Proof: We use induction on k.

Base: If $k = 2$ then this is just the previous theorem, so the base case holds. If $k = 1$, then $p|a_1$ so the base case holds.

IS: Now, suppose that for some $k \geq 2$, if $p|a_1a_2 \cdots a_k$ then there exists some i such that $p|a_i$. We want to show that if $p|a_1a_2 \cdots a_{k+1}$ then there exists some i such that $p|a_i$. If $p|a_{k+1}$, then we are done. If not, then since $p|a_1a_2 \cdots a_{k+1}$ and $p \nmid a_{k+1}$, it follows that $p|(a_1a_2 \cdots a_k)$ by bilinearity of divisibility. By the induction hypothesis, there exists some i such that $p|a_i$. $\square$

12.2 4.2 - The Fundamental Theorem of Arithmetic

Theorem: Every integer $n \geq 2$ can be written as a product of primes in a unique way, up to the order of the factors. That is, if

$$\begin{aligned} n &= p_1p_2 \cdots p_k \\ &= q_1q_2 \cdots q_m \end{aligned}$$

where p_i, q_j are primes, then $k = m$ and there exists a permutation σ of $\{1, 2, \ldots, k\}$ such that $p_i = q_{\sigma(i)}$ for all i.

Proof: We already proved the existence of prime factorizations.

Uniqueness: First, observe that if pP is prime, then $p = p$ is a prime factorization. This is the only way to write p as a product of primes, since if

$$p = p_1p_2 \cdots p_k$$

is a prime factorization, then $p|p_1p_2 \cdots p_k$, so there exists some i such that $p|p_i$. Since p_i is prime, it follows that $p = p_i$. Thus, the only prime factorization of a prime is itself.

Now, let's use strong induction on n.

- Base Case: at $n = 2$, 2 is prime, so the only prime factorization of 2 is 2 itself, so the base case holds.
- Inductive Step: Suppose k is a number such that all n such that $2 \leq n \leq k$ have a unique prime factorization. Suppose $k + 1$ has two prime factorizations

  $$p_1p_2 \cdots p_m = q_1q_2 \cdots q_n$$

  Then, $p_1|(q_1q_2 \cdots q_n)$, so there exists some i such that $p_1|q_i$. Since q_i is prime, it follows that $p_1 = q_i$. We can cancel p_1 and q_i from both sides to get

  $$p_2 \cdots p_m = q_1 \cdots q_{i-1}q_{i+1} \cdots q_n$$

  By the induction hypothesis, the remaining factors must also be equal up to a permutation. Thus, the prime factorization of $k + 1$ is unique.

12.3 EXAM 1 CUTOFF HERE

13 Lecture 12

Exam on Friday, the 27th. Covers up to Fundamental Theorem of Arithmetic.

HW due Wednesday

Note: Often, we will write the prime factorization of an integer n as

$$n = p_1^{a_1}p_2^{a_2} \ldots p_k^{a_k}$$

where all p_i are unique, and a_i are integers strictly greater than zero (eg ≥ 1)

Sometimes, we'll write a prime factorizatin of an integer $n \geq 2$ as follows:

$$2^{n_2}3^{n_3}5^{n_5}\dots$$

which resembles an "infinite product" where each prime shows up. Then, each n_i *can* be zero. Additionally we must then have that only finitely many n_p are nonzero.

<u>Example:</u> We know that $20 = 2^2 \cdot 5$. This is equal to

$$2^2 \cdot 3^0 \cdot 5^1 \cdot 7^0 \dots$$

In this case $n_2 = 2, n_3 = 0, n_5 = 1, n_p = 0$ for primes $p \geq 7$.

<u>Remark:</u> We can write $n = 1$ this way too, with $n_p = 0$ for all p.

Proposition (4.4): Let a, b be positive integers with prime factorizations

$$a = 2^{a_2}3^{a_3}5^{a_5}\dots \quad b = 2^{b_2}3^{b_3}5^{b_5}\dots$$

Then, $a|b$ iff $a_p \leq b_p$ for every prime p.

Proof: $\Rightarrow$ If $a|b$ then $b = ac$ for some integer $c \geq 1$. Write $c = 2^{c_2}3^{c_3}5^{c_5}\dots$ Then,

$$\begin{aligned} b &= ac \\ &= (2^{a_2}3^{a_3}5^{a_5}\dots)(2^{c_2}3^{c_3}5^{c_5}\dots) \\ &= 2^{a_2+c_2}3^{a_3+c_3}5^{a_5+c_5}\dots \end{aligned}$$

By the fundamental theorem of arithmetic, the prime factorization of b is unique, so $b_p = a_p + c_p$ for every prime p. Since $c_p \geq 0$, it follows that $a_p \leq b_p$ for every prime p.

$\Leftarrow$ If $a_p \leq b_p$ for every prime p, then $b_p - a_p \geq 0$, and we can write $b = 2^{b_2}3^{b_3}5^{b_5}\dots$ as

$$\begin{aligned} b &= 2^{a_2}3^{a_3}5^{a_5}\dots(2^{b_2-a_2}3^{b_3-a_3}5^{b_5-a_5}\dots) \\ &= a(2^{b_2-a_2}3^{b_3-a_3}5^{b_5-a_5}\dots) \end{aligned}$$

Since only finitely many a_p, b_p are nonzero, it follows that only finitely many $b_p - a_p$ are nonzero, so $2^{b_2-a_2}3^{b_3-a_3}5^{b_5-a_5}\dots$ is a well defined integer. Thus, $a|b$. $\square$

Proposition (4.5): Let $a = 2^{a_2}3^{a_3}5^{a_5}\dots$ and $b = 2^{b_2}3^{b_3}5^{b_5}\dots$ be positive integers. Then, let $d_p = \min(a_p, b_p)$ for every prime p. Then, $\gcd(a, b) = 2^{d_2}3^{d_3}5^{d_5}\dots$

<u>Example:</u> If $a = 720 = 2^4 \cdot 3^2 \cdot 5$ and $b = 6400 = 2^2 \cdot 5^2 \cdot 3^2 \cdot 7 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7$

Then, $d_2 = 2, d_3 = 2, d_5 = 1, d_7 = 0$, and all other $d_p = 0$, so $\gcd(a, b) = 2^2 \cdot 3^2 \cdot 5 = 180$.

Proof: Let $d_p = \min(a_p, b_p)$ for every prime p, and let $d = 2^{d_2}3^{d_3}5^{d_5}\dots$. We will show that $d = \gcd(a, b)$. By definition, $d_p \leq a_p$, and $d_p \leq b_p$ for every prime p, so by the previous proposition, $d|a$ and $d|b$. Thus, it is a common divisor.

Let e be an arbitrary common divisor. Then, $e = 2^{e_2}3^{e_3}5^{e_5}\dots$ for some nonnegative integers e_p. Since $e|a$ and $e|b$, it follows by the previous proposition that $e_p \leq a_p$ and $e_p \leq b_p$ for every prime p.

This implies that $e_p \leq \min(a_p, b_p) = d_p$ for every prime p, so by the previous proposition, $e|d$. Additionally, by def of divisibility $e \leq d$. Thus, d is the greatest common divisor of a and b. $\square$

Definition (LCM): The least common multiple of two positive integers a, b is the smallest positive integer m such that $a|m$ and $b|m$. We denote this as $\text{lcm}(a, b)$.

Proposition (4.5): (continued) If $a = 2^{a_2}3^{a_3}5^{a_5}\dots$ and $b = 2^{b_2}3^{b_3}5^{b_5}\dots$ are positive integers, then let $m_p = \max(a_p, b_p)$ for every prime p. Then, $\text{lcm}(a, b) = 2^{m_2}3^{m_3}5^{m_5}\dots$

Proof: Let $m_p = \max(a_p, b_p)$ for every prime p, and let $m = 2^{m_2}3^{m_3}5^{m_5}\dots$. We will show that $m = \text{lcm}(a, b)$. By definition, $m_p \geq a_p$ and $m_p \geq b_p$ for every prime p, so by the previous proposition, $a|m$ and $b|m$. Thus, it is a common multiple.

Corollary: $\gcd(a, b) \cdot \text{lcm}(a, b) = a \cdot b$ for all positive integers a, b.

Proof: Let $d_p = \min(a_p, b_p)$ and $m_p = \max(a_p, b_p)$ for every prime p. Then,

$$\begin{aligned}\gcd(a, b) \cdot \text{lcm}(a, b) &= (2^{d_2}3^{d_3}5^{d_5}\dots)(2^{m_2}3^{m_3}5^{m_5}\dots) \\ &= 2^{d_2+m_2}3^{d_3+m_3}5^{d_5+m_5}\dots\end{aligned}$$

Since $d_p = \min(a_p, b_p)$ and $m_p = \max(a_p, b_p)$, it follows that $d_p + m_p = a_p + b_p$ for every prime p. Thus,

$$\begin{aligned}\gcd(a, b) \cdot \text{lcm}(a, b) &= 2^{a_2+b_2}3^{a_3+b_3}5^{a_5+b_5}\dots \\ &= (2^{a_2}3^{a_3}5^{a_5}\dots)(2^{b_2}3^{b_3}5^{b_5}\dots) \\ &= a \cdot b\end{aligned}$$

$\square$

<u>Note:</u> This also implies that

$$\text{lcm}(a, b) = \frac{a \cdot b}{\gcd(a, b)}$$

<u>Example:</u> (of Prop): If $a = 720 = 2^4 \cdot 3^2 \cdot 5$ and $b = 6400 = 2^2 \cdot 5^2 \cdot 3^2 \cdot 7 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7$, then $\text{lcm}(a, b) = 2^4 3^2 5^2 \cdot 7 = 25200$.